

MATH3303 – Lecture 22

Sums and Products of Ideals, Principal Ideals, and Quotient Rings

Review Notes

Reference: Lee, Abstract Algebra, Sections 9.1–9.2

Overview

This lecture continues the study of ideals begun in Lecture 21 and introduces quotient rings:

- **Sum and product of ideals** – $I + J$ and IJ , with the warning that $\{ij : i \in I, j \in J\}$ is *not* in general an ideal
- **Intersection of ideals** – $I \cap J$ is again an ideal
- **Principal ideals** – $(a) = \{ra : r \in R\}$ in a commutative ring
- **Every ideal in $\mathbb{Z}$ is principal** – proved directly via the Division Algorithm
- **Non-principal ideals exist** – the ideal $(2, x) \trianglelefteq \mathbb{Z}[x]$ is not principal
- **Quotient rings** – R/I as a ring under coset addition and multiplication

Source labels. The title of each named result indicates its origin: *(from lecture)*, *(from Lee, X.Y)*, or *(from lecture; cf. Lee, X.Y)*.

1 Sums and Products of Ideals

Definition 1.1: Sum of ideals (from lecture; cf. Lee, §9.1)

Let R be a commutative ring and $I, J \trianglelefteq R$. The **sum** of I and J is

$$I + J = \{i + j : i \in I, j \in J\}.$$

Definition 1.2: Product of ideals (from lecture; cf. Lee, §9.1)

Let R be a commutative ring and $I, J \trianglelefteq R$. The **product** of I and J is

$$IJ = \{i_1j_1 + i_2j_2 + \cdots + i_nj_n : i_k \in I, j_k \in J, n \in \mathbb{Z}_{>0}\}.$$

Remark 1.1: Why not just $\{ij : i \in I, j \in J\}$? (from lecture)

We *cannot* simply define $IJ = \{ij : i \in I, j \in J\}$: this set is **not closed under addition** in general. Sums of two such products need not themselves be expressible as a single product ij , so we are forced to take *finite sums* of products to obtain an ideal.

Lemma 1.1: $I + J$ and IJ are ideals (from lecture; Lee, Thms. 9.3 & 9.4)

Let R be a commutative ring and $I, J \trianglelefteq R$. Then

$$I + J \trianglelefteq R \quad \text{and} \quad IJ \trianglelefteq R.$$

Proof. (1) $I + J \trianglelefteq R$.

Since $I, J \leq (R, +)$ and $(R, +)$ is abelian, $I + J$ is a subgroup of $(R, +)$:

$$(i_1 + j_1) - (i_2 + j_2) = (i_1 - i_2) + (j_1 - j_2) \in I + J.$$

Absorption property. Let $i \in I, j \in J, r \in R$. Then

$$r(i + j) = \underbrace{ri}_{\in I} + \underbrace{rj}_{\in J} \in I + J,$$

and similarly $(i + j)r \in I + J$. Hence $I + J \trianglelefteq R$.

(2) $IJ \trianglelefteq R$.

Closure under subtraction:

$$(i_1j_1 + \cdots + i_mj_m) - (i_{m+1}j_{m+1} + \cdots + i_nj_n) = i_1j_1 + \cdots + i_mj_m + (-i_{m+1})j_{m+1} + \cdots + (-i_n)j_n \in IJ.$$

For any $r \in R$, using commutativity of R :

$$r(i_1j_1 + \cdots + i_nj_n) = (ri_1)j_1 + \cdots + (ri_n)j_n \in IJ,$$

$$(i_1j_1 + \cdots + i_nj_n)r = i_1(j_1r) + \cdots + i_n(j_nr) \in IJ.$$

Hence $IJ \trianglelefteq R$. □

Lemma 1.2: Intersection of ideals (from lecture; Lee, Exercise 9.3)

Let R be a commutative ring and $I, J \trianglelefteq R$. Then $I \cap J \trianglelefteq R$.

Remark 1.2: Relations between $I + J$, IJ , and $I \cap J$ (from Lee, §9.1)

In any commutative ring, we always have the containments

$$IJ \subseteq I \cap J \subseteq I, J \subseteq I + J.$$

The first follows from absorption: each i_kj_k lies in both I and J .

Example 1.1: When $I + J = R$ (from lecture)

If $I + J = R$, then $I \cap J = IJ$.

Sketch. The containment $IJ \subseteq I \cap J$ is automatic (Remark 1.2). Conversely, since $1 \in R = I + J$, write $1 = i_0 + j_0$ with $i_0 \in I, j_0 \in J$. For $x \in I \cap J$,

$$x = x \cdot 1 = \underbrace{xi_0}_{x \in J, i_0 \in I} + \underbrace{xj_0}_{x \in I, j_0 \in J} \in IJ.$$

2 Principal Ideals

Definition 2.1: Principal ideal (from lecture; Lee, Def. 9.2)

Let R be a commutative ring (with identity) and $a \in R$. The **principal ideal generated by a** is

$$(a) = \{ra : r \in R\}.$$

Remark 2.1: (a) is the smallest ideal containing a (from Lee, Theorem 9.5)

In a commutative ring with identity, (a) is an ideal of R , and it equals the intersection of all ideals of R containing a .

Theorem 2.1: Every ideal in $\mathbb{Z}$ is principal (from lecture; cf. Lee, Example 10.12)

Let $I \trianglelefteq \mathbb{Z}$. Then $I = (a)$ for some $a \in \mathbb{Z}$.

Proof. **Case 1:** $I = \{0\}$. Then $I = (0)$, which is principal.

Case 2: $I \neq \{0\}$. Take any $c \in I \setminus \{0\}$; then $-c \in I$ also, so I contains a positive integer. By the well-ordering principle, I contains a *smallest positive integer*; call it a .

Claim: $I = (a)$.

($\supseteq$) Since $a \in I$, by absorption $ra \in I$ for every $r \in \mathbb{Z}$. Hence $(a) \subseteq I$.

($\subseteq$) Take $x \in I$. By the **Division Algorithm**, there exist $q, r \in \mathbb{Z}$ such that

$$x = aq + r, \quad 0 \leq r < a.$$

Then $r = x - aq \in I$ (since I is closed under subtraction, and $aq \in I$ by absorption). But $0 \leq r < a$, and a is the smallest positive integer in I , so the only possibility is $r = 0$.

Thus $x = aq \in (a)$, giving $I \subseteq (a)$.

Therefore $I = (a)$. □

Exercise 2.1: Every ideal in $K[x]$ is principal (from lecture; Lee, Example 10.13)

Let K be a field. Show that every ideal of the polynomial ring $K[x]$ is principal.

Hint. Mimic the proof of Theorem 2.1, replacing the Division Algorithm for $\mathbb{Z}$ by the Division Algorithm for $K[x]$, and “smallest positive integer” by “nonzero polynomial of smallest degree”.

3 Ideals Generated by Several Elements

Definition 3.1: Ideal generated by $a_1, \dots, a_n$ (from lecture; cf. Lee, §9.1)

Let R be a commutative ring (with identity) and $a_1, a_2, \dots, a_n \in R$. The **ideal generated by** $a_1, \dots, a_n$ is

$$(a_1, a_2, \dots, a_n) = \sum_{k=1}^n (a_k) = \{ r_1 a_1 + r_2 a_2 + \dots + r_n a_n : r_i \in R \}.$$

Example 3.1: A non-principal ideal in $\mathbb{Z}[x]$ (from lecture; Lee, Example 10.14)

The ideal $I = (2, x) \leq \mathbb{Z}[x]$ is **not** principal.

Proof. Note first that

$$I = (2, x) = \{ 2f(x) + xg(x) : f, g \in \mathbb{Z}[x] \} = \{ h(x) \in \mathbb{Z}[x] : \text{constant term of } h \text{ is even} \}.$$

Suppose for contradiction that $I = (f(x))$ for some $f(x) \in \mathbb{Z}[x]$. Since $2 \in I$, we have $2 = f(x)g(x)$ for some $g(x) \in \mathbb{Z}[x]$. As $\mathbb{Z}[x]$ is an integral domain, comparing degrees forces $f(x), g(x)$ to be constant, so $f(x) \in \{\pm 1, \pm 2\}$.

If $f(x) = \pm 1$, then $(f(x)) = \mathbb{Z}[x]$, but $1 \notin I$ (the constant term 1 is odd). Contradiction. Hence $f(x) = \pm 2$, so $(f(x)) = 2\mathbb{Z}[x]$.

But $x \in I$, so $x \in (f(x)) = 2\mathbb{Z}[x]$, meaning $x = 2h(x)$ for some $h(x) \in \mathbb{Z}[x]$ – i.e. $2 \mid x$ in $\mathbb{Z}[x]$. This is absurd (the coefficient of x on the right is even, on the left is 1).

Therefore $(2, x)$ is not principal. □

Remark 3.1: $\mathbb{Z}[x]$ is not a PID (from Lee, Example 10.14)

A commutative ring with identity in which every ideal is principal is called a **principal ideal domain (PID)**. Theorem 2.1 says $\mathbb{Z}$ is a PID, and Exercise 2.1 says $K[x]$ is a PID for K a field. Example 3.1 shows $\mathbb{Z}[x]$ is *not* a PID.

4 Quotient Rings

Remark 4.1: Cosets of an ideal (from lecture; cf. Lee, §9.2)

Let R be a commutative ring and $I \trianglelefteq R$. Since $(R, +)$ is abelian, I is automatically a normal subgroup of $(R, +)$. The **left coset** of I in R with representative $r \in R$ is

$$r + I = \{r + i : i \in I\}.$$

The set of cosets $R/I = \{r + I : r \in R\}$ forms an abelian group under

$$(a + I) + (b + I) = (a + b) + I.$$

Recall that $a + I = b + I$ if and only if $a - b \in I$.

Definition 4.1: Quotient ring (from lecture; Lee, Def. 9.3)

Let R be a commutative ring and $I \trianglelefteq R$. The **quotient ring** (or **factor ring**) R/I is the set of all left cosets $\{r + I : r \in R\}$, equipped with

$$(a + I) + (b + I) = (a + b) + I,$$

$$(a + I)(b + I) = ab + I.$$

Theorem 4.1: R/I is a ring (from lecture; Lee, Theorem 9.6)

Let R be a commutative ring and $I \trianglelefteq R$. Then $(R/I, +, \cdot)$ is a ring.

Proof. From Remark 4.1, $(R/I, +)$ is an abelian group.

Multiplication is well-defined. Suppose $a_1 + I = a_2 + I$ and $b_1 + I = b_2 + I$. Then $a_1 - a_2 \in I$ and $b_1 - b_2 \in I$. We compute

$$a_1b_1 - a_2b_2 = a_1b_1 - a_1b_2 + a_1b_2 - a_2b_2 = a_1(b_1 - b_2) + (a_1 - a_2)b_2.$$

By absorption, both $a_1(b_1 - b_2) \in I$ and $(a_1 - a_2)b_2 \in I$, so $a_1b_1 - a_2b_2 \in I$. Hence $a_1b_1 + I = a_2b_2 + I$.

Associativity of multiplication. Inherited from associativity in R :

$$((a + I)(b + I))(c + I) = (ab + I)(c + I) = (ab)c + I = a(bc) + I = (a + I)((b + I)(c + I)).$$

So $(R/I, \cdot)$ is a monoid (with identity $1 + I$ when R has identity).

Distributive laws. Also inherited from R :

$$(a + I)((b + I) + (c + I)) = (a + I)(b + c + I) = a(b + c) + I = (ab + ac) + I = (a + I)(b + I) + (a + I)(c + I).$$

□

Example 4.1: $\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}_n$ (from Lee, Example 9.24)

For $n \geq 2$, the quotient $\mathbb{Z}/n\mathbb{Z}$ is precisely the ring $\mathbb{Z}_n$ of integers modulo n . The cosets are

$$0 + n\mathbb{Z}, \quad 1 + n\mathbb{Z}, \quad 2 + n\mathbb{Z}, \quad \dots, \quad (n - 1) + n\mathbb{Z},$$

which we usually write as $[0], [1], \dots, [n-1]$.

Checklist of skills

- ☐ State the definitions of $I + J$ and IJ , and explain why $IJ \neq \{ij : i \in I, j \in J\}$.
- ☐ Prove that $I + J$, IJ , and $I \cap J$ are ideals.
- ☐ State the principal ideal (a) and the ideal $(a_1, \dots, a_n)$.
- ☐ Prove that every ideal in $\mathbb{Z}$ is principal using the Division Algorithm.
- ☐ Prove that every ideal in $K[x]$ is principal.
- ☐ Show that $(2, x) \trianglelefteq \mathbb{Z}[x]$ is not principal.
- ☐ Define cosets of an ideal and the quotient ring R/I .
- ☐ Verify that multiplication on R/I is well-defined.
- ☐ Prove that R/I is a ring.